

Consumer Choice II

Alejandro Melo Ponce



Stony Brook University

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The budget constraint

- A budget constraint is a constraint on how much money (income, wealth) an agent can spend on goods.
- Let $I \geq 0$ denote available income. $(x, y) \in \mathbb{R}_+^2$ denotes a consumption bundle.
- We assume that we can observe the prices of the goods (p_x, p_y)
- The **budget set** is then described by

$$\{(x, y) \in \mathbb{R}_+^2 \mid p_x x + p_y y \leq I\}$$

- Expenditures on x : $p_x x$, Expenditures on y : $p_y y$. The sum of both cannot exceed I .
- The **budget line** is: $p_x x + p_y y = I$.
- Slope of the budget line $\frac{dy}{dx} = -\frac{p_x}{p_y}$

Changes in the budget line

- Write the budget line as

$$y = \frac{I}{p_y} - \frac{p_x}{p_y}x$$

- Notice that the intercepts are I/p_y and I/p_x .
- Then it is easy to see what happens when we change either prices or income.
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Taxes and subsidies

- With this framework, is easy to analyze the impact of subsidies and taxes on the consumer's budget constraint.
- Taxes
 - Quantity tax: The consumer has to pay certain amount for each unit of the good she purchases.
 - A quantity tax of t dollars on good x increases its price from p_x to $p_x + t$
 - Ad valorem tax: The tax is on the value, i.e. price of the good. If the tax rate on good x is τ then the actual price that the consumer faces is $(1 + \tau)p_x$
- Similarly for subsidies:
 - Quantity subsidy: For good x then the price the consumer faces for a quantity subsidy of s per unit is $p_x - s$.
 - Ad valorem subsidy: If the rate for subsidy of good x is σ then the price becomes $(1 - \sigma)p_x$.

The consumers Utility Maximization Problem (UMP)

- We will model consumer choice **as if** he were maximizing utility.
- Do economists think that consumers actually maximize utility? **NO**.
- However, the model will yield some testable predictions that will actually conform to reality.
- Is this a good model? Depends on what you are analyzing.
- Advantage: wide range of applications under the same fundamental mathematical model.

Definition (Utility Maximization Problem)

Suppose that preferences over bundles $(x, y) \in \mathbb{R}_+^2$ are represented by $u(x, y)$, prices are (p_x, p_y) and income is I

$$\begin{array}{ll} \text{maximize} & u(x, y) \\ \text{subject to} & p_x x + p_y y \leq I \end{array} \quad (\text{UMP})$$

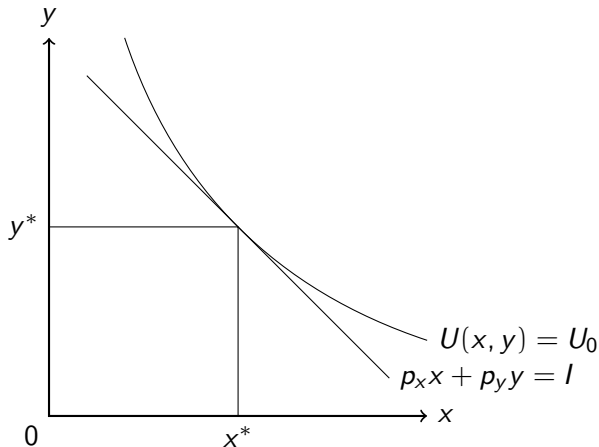
Graphical Analysis: Intuition

At an **interior** optimum, the slopes of the utility function and the budget line need to be equal. Thus

$$-\frac{p_x}{p_y} = -\frac{u_x}{u_y}$$

$$\frac{p_x}{p_y} = \frac{u_x}{u_y}$$

$$\frac{p_x}{p_y} = MRS$$



A detour on Unconstrained Optimization

Let f be a real-valued function defined over $\mathbb{R}^n$, that means $f : \mathbb{R}^n \rightarrow \mathbb{R}$.

Definition

The function f has a **local maximum** at x^* if there exists a neighborhood I such that it contains x^* and $f(x^*) \geq f(x)$ for all $x \in I$.

Definition

The function f has a **global maximum** at x^* if $f(x^*) \geq f(x)$ for all x in the domain of f .

First and Second order conditions: Single variable case

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a real-valued single variable function.

Theorem (First Order Necessary Condition for maximum/minimum)

Let f be a real-valued function of one variable. Suppose that f has a local maximum/minimum at some point x^ and suppose that $f'(x)$ exists. Then $f'(x^*) = 0$.*

Theorem (Second Order Sufficient condition for a maximum(minimum))

Let f be a real-valued function of one variable. Suppose that $f''(x^)$ exists and suppose that $f'(x^*) = 0$ and $f''(x^*) < 0$ ($f''(x^*) > 0$). Then f has a local maximum(minimum) at x^* .*

Multivariate function Unconstrained maximization

Theorem (First Order Necessary Condition for maximum/minimum)

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$. Suppose that f has a local maximum/minimum at some point x^* and suppose that f is differentiable. Then $\frac{\partial f(x_1, \dots, x_n)}{\partial x_i} = 0$ for all $i = 1, \dots, n$

- To get second order sufficient conditions for maxima (minima), we need to be able to talk about when a multivariate function f is concave (convex).
- We will derive conditions on second order partial derivatives for this to happen. But first...

Theorem (Young's Theorem)

Suppose that $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a real-valued multivariate function that has *continuous second order partial derivatives*. Then $f_{ij} = f_{ji}$ for $i \neq j$, $i, j = 1, 2, \dots, n$.

Second order conditions

- We will consider the case of a real-valued function of two variables $f(x, y)$.
- What conditions must we put on the partial derivatives to ensure that the change in f is negative when we move away from the critical point?
- The total differential is

$$df = f_x dx + f_y dy \quad (1)$$

- The total differential of (1) is

$$\begin{aligned} d^2f &= (f_{xx} dx + f_{xy} dy) dx + (f_{yx} dx + f_{yy} dy) dy \\ &= f_{xx} dx^2 + 2f_{xy} dx dy + f_{yy} dy^2 \quad \text{By Young's Theorem} \end{aligned} \quad (2)$$

- We want for d^2f to be unambiguously negative.
 - Suppose $dy = 0$, $d^2f < 0 \Rightarrow f_{xx} < 0$
 - Suppose $dx = 0$, $d^2f < 0 \Rightarrow f_{yy} < 0$
 - Suppose neither dx or dy are zero. Then we need $f_{xx}f_{yy} - f_{xy}^2 > 0$.

Back to constrained Optimization

Definition (First order conditions for interior maximum)

Suppose preferences are represented by the utility function $u(x, y)$. If (x^*, y^*) is an interior constrained maximum, then we must have

$$\frac{p_x}{p_y} = MRS(x^*, y^*)$$

Definition (Second order conditions for interior maximum)

A decreasing MRS is sufficient to guarantee that the point identified by the FOC $MRS = p_x/p_y$ is indeed the constrained maximum. What conditions on the utility function guarantee a decreasing MRS? If preferences are convex we have seen that they can be represented with a utility function which will have a decreasing MRS. What we are looking for is on conditions for the derivatives of u .

Optimization by Substitution

- Suppose that the consumer has **monotonic preferences**, i.e. the more the better.
- Then we know for sure that he will try to exhaust his income.
- Thus, the restriction of the budget constraint will hold with equality:
 $p_x x + p_y y = I$.
- Thus we can substitute $y = \frac{I - p_x x}{p_y}$ into the utility function to get

$$u(x, y) = u\left(x, \frac{I - p_x x}{p_y}\right)$$

- We can treat this problem as a single variable unconstrained maximization problem:

$$\begin{aligned}\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} \frac{dy(x)}{dx} &= 0 \\ \frac{\partial u}{\partial x} - \frac{\partial u}{\partial y} \frac{p_x}{p_y} &= 0 \\ \Rightarrow \frac{p_x}{p_y} &= \frac{\frac{\partial u}{\partial x}}{\frac{\partial u}{\partial y}} = MRS\end{aligned}$$

- What about second order conditions?

What can go wrong? Corner solutions, Bliss points

- The condition $MRS = p_x/p_y$ helps identify an interior optimum.
- What happens when it is optimal to consume positive quantities of one good, but none of the other?
- Examples, perfect substitutes.
- On the other hand, what if the preference do not satisfy the more the better?
- If preference have a **bliss point** it is possible that the optimum can happen **inside the budget set**.
- We need to amend the first order condition to deal with these cases.